PROJECT-3 REPORT

on

Sarthak Nivesh
(Financial Market Analysis)

Submitted by:

Aman Jain- 230572
Rohit Fogla- 230597
Vanshita Mehta - 230734
Dishita Tirthani-230604

under mentorship of

Dr. Yogesh Gupta

(Professor)

Department of Computer Science Engineering

School of Engineering and Technology

BML MUNJAL UNIVERSITY, GURUGRAM (INDIA)

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Candidate Declaration

What we are about to state is that the project under the title of Sarthak Nivesh - AI Powered Financial Market Investment Intelligence Platform is a one of a kind of work done by us as a part of our academic program. The project has been developed by the group members:

Aman Jain

Rohit Fogla

Dishita Tirthani

Vanshita Mehta

under the priceless advice and guidance of Dr. Yogesh Gupta.

We also state that this project has not been previously, either in whole or in part, submitted to the granting of any degree, diploma, or any other educational award in this or any other institution. References and all information and data utilized in this project have been given due credit.

The project pertains to the field of a financial market and is aimed at the creation of an artificial intelligence-driven platform, which will offer Indian retail investors stock analysis, IPO insights, and mutual funds comparative analysis, an intelligent money tracker, and investment advice called Sarthak Nivesh.

We also guarantee the originality and genuineness of the work in this report.

Name	**Signature**
Aman Jain	
Rohit Fogla	
Dishita Tirthani	
Vanshita Mehta	

Date: ____________
Place: ____________

Supervisor Declaration

This is to certify that the project entitled **"Sarthak Nivesh – AI Powered Financial Market Investment Intelligence Platform"** is a bonafide work carried out by:

- Aman Jain
- Rohit Fogla
- Dishita Tirthani
- Vanshita Mehta

during the academic session under my supervision and guidance.

The students have successfully completed the project in the field of financial market analysis and investment intelligence. The work presented in this report demonstrates their sincere effort, technical understanding, and practical implementation skills. The project focuses on developing an integrated platform for stock analysis, IPO intelligence, mutual fund comparison, smart money tracking, portfolio management, and AI-based investment recommendations for Indian retail investors.

I further certify that the work embodied in this project report is original and has been completed by the above-mentioned students. To the best of my knowledge, this project has not been submitted earlier, either in full or in part, to any university or institution for the award of any degree, diploma, or other academic qualification.

The project fulfills the requirements prescribed for the course, and I recommend it for submission and evaluation.

Dr. Yogesh Gupta
Project Guide / Supervisor

Acknowledgement

We are pleased to acknowledge that we have had a valuable experience of being supported, guided and encouraged by our respected mentor and guide, Dr. Yogesh Gupta in terms of the development of our project, Sarthak Nivesh, which is an AI Powered financial market Investment Intelligence Platform. His expert advice and motivation helped us in understanding the concepts of financial markets, artificial intelligence, and project implementation in a much better way.

We are grateful to our department and institution which offered us the opportunity, resources and environment needed to accomplish this project successfully. We also wish to credit all the faculty members who directly/indirectly assisted us in the project work.

We would like to state that we are immensely grateful to our team members Aman Jain, Rohit Fogla, Dishita Tirthani and Vanshita Mehta with regard to their dedication, joint efforts, and constant efforts towards delivering this project. All members had their own areas of contribution that include stock analysis, smart money tracking, IPO intelligence, portfolio management, AI integration, mutual fund analysis and user interface design.

There are also the developers and organizations of the different technologies and sources of data that we utilized in our project, such as Python, Streamlit, Groq AI, Yahoo Finance, NSE APIs, Google News RSS, AMFI, and other open-source libraries that we would like to thank. They had tools and platforms that were significant in making this project possible.

Lastly, we owe our family and friends an in-depth enthusiasm, patience, and moral support throughout the project.

This has been a learning experience for us in this project and we are very grateful to everyone who played their part in ensuring that this project was successfully completed.

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Abstract

“Sarthak Nivesh” is an AI-powered financial market investment intelligence platform developed specifically for Indian retail investors. The main objective of the project is to provide a single, free, and intelligent platform that helps users make informed investment decisions related to stocks, IPOs, mutual funds, and portfolio management. At present, investors need to depend on multiple websites and expensive tools to gather market information, which makes the process difficult and time-consuming. This project aims to solve that problem by integrating all important financial services into one unified platform.

The platform uses real-time financial data collected from reliable sources such as Yahoo Finance, NSE APIs, Google News RSS, AMFI, and IPO-related websites. It analyzes more than 120 Indian stocks using 50+ technical indicators, institutional money flow, market sentiment, and company fundamentals. Based on this analysis, the system provides BUY, SELL, or HOLD recommendations along with confidence score, entry price, target price, stop-loss, and detailed reasoning. The project also includes features such as Smart Money Tracker, IPO Intelligence Hub, Mutual Fund Center, Portfolio Management, and News Sentiment Analysis.

A major highlight of the project is the use of Agentic AI, where multiple AI agents work together to generate better and more accurate investment recommendations. One agent performs technical analysis, another tracks FII/DII activities, the third evaluates market risk, and the final agent generates the overall strategy. The project is implemented using Python, Streamlit, Groq AI, and SQLite, making it efficient, fast, and easy to use. Unlike many existing systems, the platform follows a zero hardcoded data approach and provides only live and updated information.

The proposed system is expected to improve financial literacy and reduce the chances of loss among retail investors by providing reliable and explainable investment guidance. By combining advanced analytics, artificial intelligence, and real-time financial data into a single application, “Sarthak Nivesh” offers an affordable and accessible alternative to expensive professional investment platforms.

Chapter 1 - Introduction

Over the past years, the Indian stock market has been experiencing a high rate of growth with over 10 crores demat accounts being registered in the country. This growth notwithstanding, a great percentage of retail investors are still losing their money due to lack of appropriate knowledge, tools and advice to make wise investment decisions. Majority of new investors use social media, YouTube video, WhatsApp alternate tips or unverified market rumors rather than analysis based on data. This has made almost 95 percent of retail investors to make a low level of consistent returns in the stock market.

Meanwhile, institutional and professional investors can use advanced analytical tools at the same time, including the Bloomberg Terminal, Reuters, as well as other financial portals that are paid. These applications have technical signals, institutional capital movement, portfolio management, IPO tracking and market forecasts. Nonetheless, these platforms are very costly, as the subscription fees vary between ₹1,500 per month to almost 200,000 per year, so they are not affordable to regular investors. Hence an enormous disjuncture between available resources to institutional and common retail investors exists.

The other significant problem that is experienced by investors is the fragmentation of the financial information. A single investment decision can require an investor to go to a variety of websites, e.g. NSE India and institutional flow, Moneycontrol and news, Screener.in and fundamentals, trading sites to get charts and individual portals to get IPO and mutual fund data. This disjointed strategy is time consuming and in most cases misleading to investors since each of the platforms offers a dissimilar type of information with no singular explanation to the same.

To fix such issues, the project Sarthak Nivesh is created as the application of AI as an Indian Stock Market Investment Intelligence Platform. The name of the platform is Sarthak Nivesh, or Meaningful Investment and the aim of the platform is to equip smaller-sized retail investors with the same capability to analyze the markets as the professional traders and institutions do. It is free and offers all crucial services associated with investments in a single application.

The platform is developed with the new technologies, including Python, Streamlit, Groq AI, Yahoo Finance, and NSE APIs. It gathers live price of stocks, institutional flow data, news about the market, IPO subscription, and mutual fund NAVs of credible and official sources. This project has the advantage of having a zero hardcoded data methodology unlike many of the currently available applications that utilize their hardcoded or even static data, where all information is dynamically loaded when shown to the user regardless of whether it is a text or graphic display. This will make sure that the information is up-to-date, precise and accurate.

The Stock Intelligence Hub which analyzes over 120 Indian stocks using over 50 technical indicators like RSI, MACD, Moving Averages, Bollinger Bands, and volume analysis is one of the most significant aspects of the project. Upon the evaluation of these indicators and the company fundamentals and market sentiment, the system will give a final recommendation of either BUY, SELL, or HOLD. It also contains recommended entry point, target point, stop-loss point, confidence percentage, reasons behind the recommendation as well as risk involved.

The project is also tied with a series of other modules namely; Smart Money Tracker, IPO Intelligence Hub, Mutual Fund Center, News Sentiment Analysis, Portfolio Management, and Agentic AI Hub. The Smart Money Tracker tracks selling and buying trends of Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) thus enabling investors to know where big money is flowing in the market. In the same vein, IPO Intelligence Hub predicts the choice on whether to apply to or avoid an IPO based on the analysis of the Grey Market Premium, subscription status and investor demand.

The second element of the project that is unique is the application of Agentic AI. The system has four tailored AI agents, instead of relying on one specific AI model, to collaborate. A single agent conducts technical analysis, another agent will monitor institutional activity, the third agent will carry out the assessment of risk and finally another agent will integrate all the information to present a complete strategy. This multi-agent system enhances forecasts and makes the platform smarter and reliable as compared to traditional financial tools.

Finally, Sarhk Nivesh is not only a stock market application, but also an entire investment intelligence platform targeted towards Indian retail investors. It tries to lessen the losses, enhance financial literacy, and democratize the access to quality financial analysis. Seamlessly fusing real-time information, artificial intelligence, stock analysis, IPO monitoring, mutual funds and portfolio management under a single platform, the project will be able to make investment choices more substantial, knowledgeable and reachable to everyone.

Chapter 2 - Literature Review

The stock market prediction domain has witnessed significant advancements with the integration of machine learning, deep learning, and artificial intelligence techniques. Earlier approaches primarily relied on statistical and time-series models, but recent research has shifted towards hybrid systems that combine technical indicators, sentiment analysis, and AI-driven decision-making. These advancements have improved prediction accuracy and enabled better understanding of complex market behavior. However, due to the dynamic and uncertain nature of financial markets, achieving consistent and reliable predictions remains a challenging task.

In recent years, researchers have emphasized the importance of combining multiple data sources such as historical price data, financial indicators, and news sentiment to enhance forecasting performance. Deep learning models like LSTM and GRU have shown strong capabilities in capturing temporal dependencies, while sentiment analysis has contributed to understanding market psychology. Despite these improvements, most existing systems focus on isolated aspects of analysis rather than providing a unified, real-time, and explainable investment solution. This gap forms the basis for the development of the proposed system.

Table 2.1 : Summary of Research papers

S.No	**Year**	**Research Paper Title**	**Methodology / Approach**	**Key Findings**	**Limitations**
1	2008	AZFinText System (Foundation Paper)	Text mining + financial news analysis	Demonstrated that news sentiment impacts stock prices	Limited to textual data only
2	2020	Stock Price Prediction Using LSTM	Deep Learning (LSTM)	High accuracy in time-series prediction	Ignores external factors like sentiment
3	2021	LSTM with Sentiment Analysis	LSTM + NLP (BERT/Text)	Improved prediction accuracy using sentiment	Complex model implementation
4	2022	Stock Prediction using Deep Learning & Sentiment	LSTM + sentiment features	Multi-factor models outperform single models	Requires large datasets
5	2021	Machine Learning for Indian Stock Market	ANN, SVM, ML models	AI models better than traditional methods	Limited real-time implementation
6	2020	Optimized LSTM for Time Series	LSTM optimization techniques	Better accuracy with parameter tuning	Computationally expensive

S.No	Year	Research Paper Title	Methodology / Approach	Key Findings	Limitations
7	2025	Hybrid LSTM-ARIMA Model	Combination of statistical + deep learning	Captures both linear and nonlinear trends	Complex model integration
8	2025	AI in Financial Market Prediction	Review of AI models	AI improves forecasting performance significantly	Lacks unified system approach
9	2023	News Sentiment and Market Dynamics	Sentiment analysis + correlation study	News sentiment affects short-term trends	Not sufficient alone
10	2024	Comparative Study of Deep Learning Models	RNN, LSTM, GRU, CNN	Model selection impacts performance	No real-time deployment
11	2024	Hybrid RNN-LSTM with Sentiment	Hybrid deep learning + NLP	Reduced prediction error significantly	High complexity
12	2024	LSTM vs GRU Comparative Study	Deep learning comparison	GRU faster, LSTM more accurate	Trade-off between speed and accuracy

2.2 Research Gaps

Although significant progress has been made in stock market prediction using machine learning and deep learning techniques, several limitations still exist in current research. Most studies focus on a single dimension such as technical analysis, sentiment analysis, or time-series forecasting, without integrating multiple factors into a unified system. Additionally, many models lack explainability, making it difficult for users to understand the reasoning behind predictions. Real-time implementation is another major challenge, as most research is conducted on historical datasets rather than live market data.

Furthermore, existing studies do not adequately address aspects such as institutional investment tracking (FII/DII), IPO analysis, and portfolio-level risk management. There is also a lack of user-friendly platforms that can deliver actionable insights to retail investors in a simplified manner. The proposed *Sarthak Nivesh* system addresses these gaps by integrating multi-source real-time data, combining multiple analytical techniques, and implementing an Agentic AI-based decision-making framework.

Chapter 3 - Problem Statement

Retail investors have continued to grapple with making profitable and informed investment decisions, with the Indian stock market gaining a lot of popularity. Almost 95 percent of retail investors lose their money due to investing according to social media tips, YouTube videos or tips issued by friends instead of doing a proper financial analysis. They lack access to sophisticated analytical tools, real-time information, and expert advice in contrast to institutional investors. Market intelligence in professional sites like Bloomberg and Reuters has all the features, but is prohibitively expensive, at 1500 a month or nearly 200000 a year. This renders them unaffordable to the average investors.

The second issue is another significant one, as all financial data is spread across the websites. To find the FII/DII data, market news, and company fundamentals, investors have to use other sites, NSE India, Moneycontrol, Screener.in, and similar sites to find IPOs, mutual funds or technical charts. This disjointed experience is slow and baffling, and can result in making bad choices. Further, the majority of the free tools lack AI-generation of recommendations, score of confidence, or even the rationale of investment choice. Hence, there is an acute necessity of the existence of the unified, free, and AI powered platform that has the capability to deliver reliable stock and IPO, mutual fund and market analysis in a single location.

- Almost 95 percent of retail investors make losses because they did not get guidance and analysis.
- Unverified tips, social media and YouTube videos have taken the place of data-driven decision-making by investors.
- Common investors are not able to afford their professional financial tools like Bloomberg.
- The key information about investments is divided on various sites and apps.
- The majority of the existing free platforms lack the marriage of stocks, IPOs, mutual funds, smart money tracker, and AI recommendations.
- Not all the tools offer recommendations, including BUY/SELL/HOLD and entry price, target, and stop-loss, and confidence score are explainable.
- What institutional investors can do easily as retail investors can hardly do is to monitor institutional activities i.e., FII/DII flow, bulk dealings and block deals.
- No fully free AI-driven platform with Indian stock market paru ticularly in mind exists.

Chapter 4 - Methodology

4.1 Overview of Methodology

The proposed Agentic AI Investment Hub follows a structured and modular methodology that integrates multiple data sources, analytical agents, and artificial intelligence techniques to generate intelligent investment recommendations. The system is designed using a multi-agent architecture, where each agent performs a specialized analytical task, and the outputs are synthesized using a Large Language Model (LLaMA 3.3 70B). The methodology begins with data acquisition, followed by preprocessing, feature engineering, parallel agent-based analysis, LLM-based synthesis, and finally visualization and decision support. This approach ensures scalability, accuracy, and real-time responsiveness in financial decision-making.

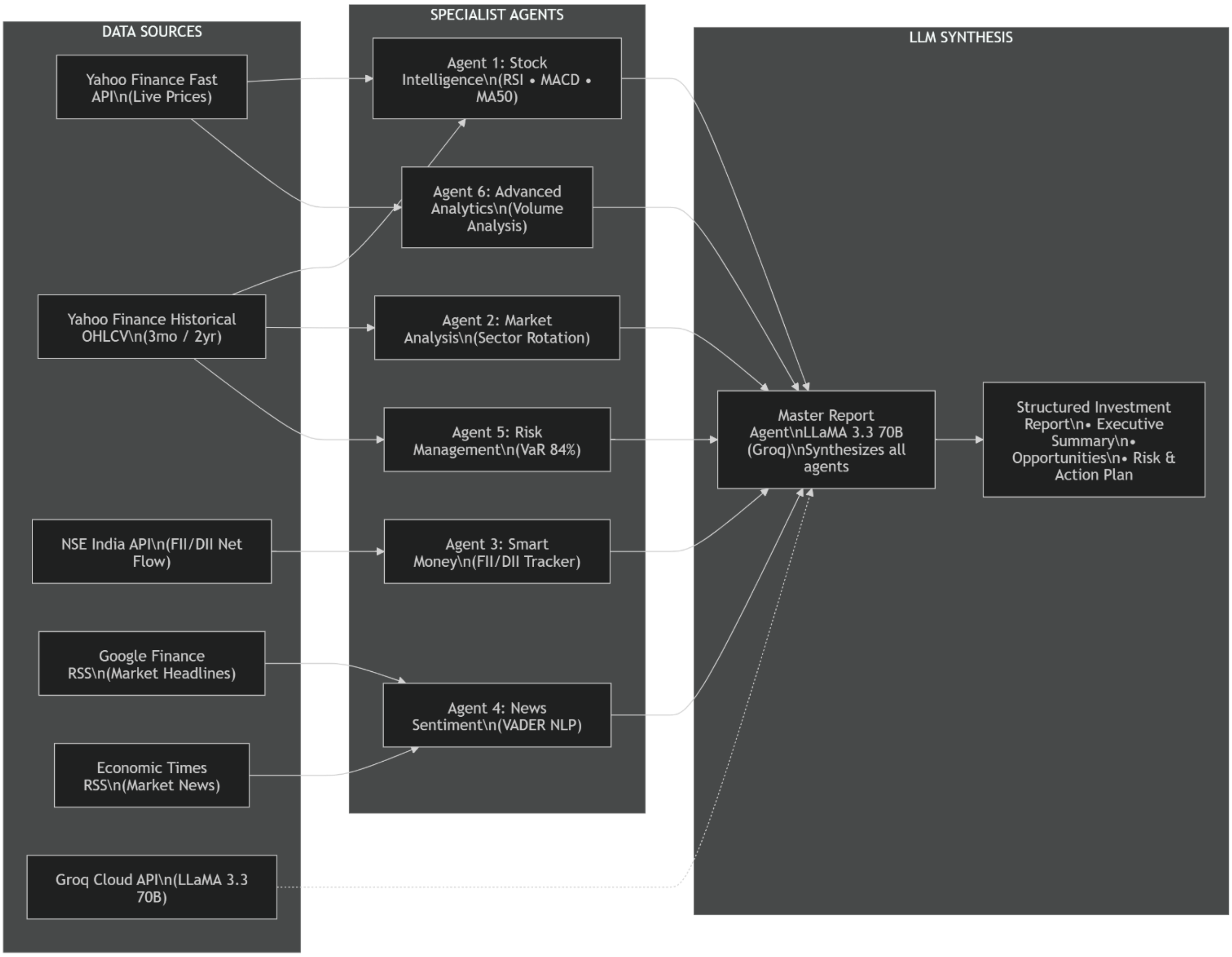

Fig 4.1 – Working of Agentic analysis

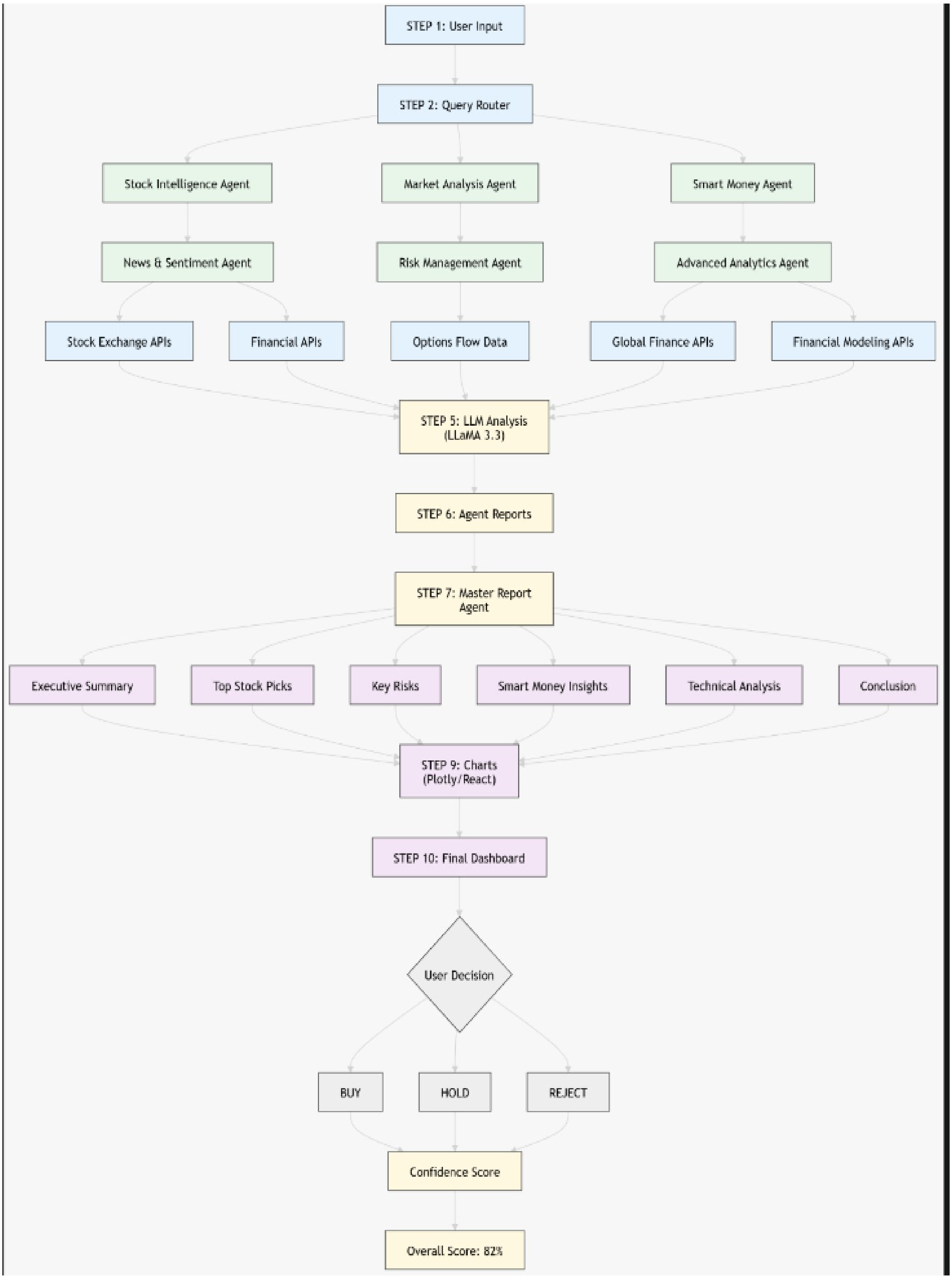

Fig 4.2 - Workflow analysis

4.2 Data Collection (Dataset Layer)

The first step in the methodology involves collecting diverse financial datasets from multiple reliable sources to ensure comprehensive market coverage. The system utilizes both structured and unstructured data. Structured data includes real-time stock prices and historical OHLCV (Open, High, Low, Close, Volume) data obtained from Yahoo Finance APIs, as well as institutional investment data such as Foreign Institutional Investors (FII) and Domestic Institutional Investors (DII) flows sourced from NSE India APIs. Additionally, volume and trading activity data are also collected for advanced analytics.

Unstructured data is obtained from financial news sources such as Google Finance RSS feeds and Economic Times RSS feeds, which provide real-time market headlines and news articles. These textual datasets are essential for sentiment analysis and event detection. Furthermore, the system integrates with the Groq Cloud API to access the LLaMA 3.3 70B model for advanced reasoning and report generation. The dataset includes both real-time streaming data and historical records spanning multiple years, enabling comprehensive short-term and long-term analysis.

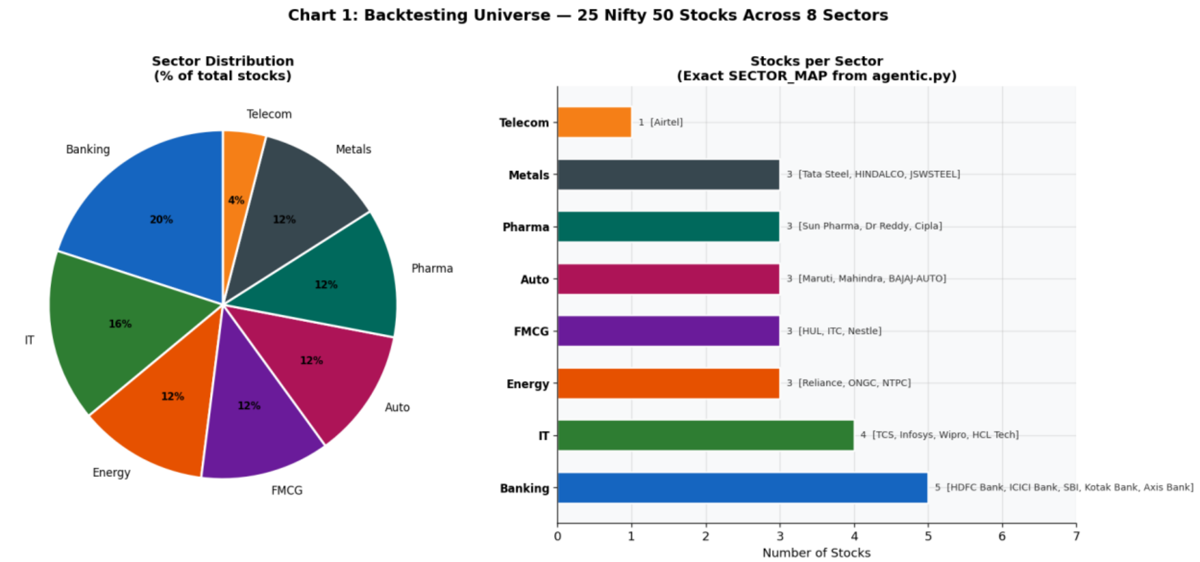

Fig 4.3 –Dataset into Sector Distribution in stocks

4.3 Data Preprocessing and Cleaning

After data collection, the raw datasets undergo preprocessing and cleaning to ensure accuracy, consistency, and usability. For structured data, missing values are handled using interpolation and filtering techniques, and the data is normalized to maintain uniformity across different sources. Time-series alignment is performed to synchronize datasets collected from different APIs.

For unstructured data, Natural Language Processing techniques are applied. This includes removal of stopwords, punctuation, and irrelevant symbols, followed by tokenization and lemmatization. Duplicate and redundant news entries are also filtered out to reduce noise. These preprocessing steps ensure that the dataset becomes clean, structured, and suitable for further analysis, thereby improving the reliability of the system.

4.4 Feature Engineering

Feature engineering is performed to transform raw data into meaningful analytical inputs. Several technical indicators are computed from historical OHLCV data, including the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Moving Average (MA50). These indicators help identify market trends, overbought and oversold conditions, and potential trend reversals.

In addition, market-related features such as sector performance, institutional activity (FII/DII flows), and volume anomalies are extracted. Sentiment features are generated by assigning polarity scores to news data using the VADER sentiment analysis model. These engineered features provide a comprehensive representation of market behavior and significantly enhance predictive performance.

4.5 Query Processing and Routing

The system accepts user queries related to investment decisions, such as stock selection, sector analysis, or long-term investment planning. These queries are processed using Natural Language Processing techniques to extract user intent and determine the required analytical tasks. A Query Router then directs the request to appropriate agents based on the type of analysis needed, the data sources required, and the complexity of the query. This dynamic routing mechanism ensures efficient processing and optimal utilization of system resources.

4.6 Multi-Agent Analytical Pipeline

The figure illustrates the complete architecture of the proposed Agentic AI Investment Hub, highlighting the flow of data from multiple sources through specialized analytical agents to the final decision-making stage. On the left, the system collects data from various structured and unstructured sources, including live and historical stock prices from Yahoo Finance, institutional flow data from NSE (FII/DII), and financial news from Google Finance and Economic Times RSS feeds. Additionally, the Groq Cloud API provides access to the LLaMA 3.3 70B model, which is used for advanced reasoning and report generation. This diverse data integration ensures that the system captures real-time market dynamics as well as historical trends.

In the central layer, the architecture employs multiple specialized agents that operate in parallel, each focusing on a specific analytical domain. The Stock Intelligence Agent uses technical indicators such as RSI, MACD, and moving averages to generate trading signals, while the Market Analysis Agent identifies sector rotation trends. The Smart Money Agent tracks institutional investments, and the News Sentiment Agent analyzes market sentiment using VADER NLP. Additionally, the Risk Management Agent evaluates risk through metrics like Value at Risk (VaR), and the Advanced Analytics Agent detects volume anomalies and unusual trading patterns. This multi-agent design enables comprehensive and domain-specific analysis, improving the overall accuracy and robustness of the system.

The final stage of the architecture is the LLM synthesis layer, where all agent outputs are aggregated by the Master Report Agent powered by LLaMA 3.3 70B. This component synthesizes insights from different agents, resolves conflicting signals, and generates a structured investment report. The final output includes an executive summary, identified opportunities, and a risk and action plan, providing users with clear and actionable recommendations. Overall, the architecture demonstrates a seamless integration of data processing, parallel AI agents, and advanced language models to deliver an intelligent and explainable investment decision-support system.

4.7 Specialized Investment Modules

In addition to the core agents, the system includes specialized modules that extend its capabilities.

The SIP Planner simulates systematic investment strategies by modeling periodic investments, enabling evaluation of long-term portfolio growth and stability. The IPO Intelligence Hub analyzes new market offerings using subscription data, sentiment analysis, and company fundamentals to provide Apply or Avoid recommendations. The Mutual Funds Intelligence Module evaluates mutual fund performance based on historical returns, risk metrics, and diversification strategies, offering low-risk investment alternatives.

These modules enhance the system by covering a wide range of investment strategies, from short-term trading to long-term financial planning.

4.8 LLM-Based Synthesis (Core Intelligence Layer)

The outputs generated by all agents are passed to the Master Report Agent, powered by the LLaMA 3.3 70B model. This component acts as the central intelligence layer of the system. It aggregates insights from different agents, resolves conflicting signals, and performs contextual reasoning to generate meaningful conclusions.

The LLM converts complex analytical outputs into structured and human-readable reports. It ensures coherence, logical consistency, and interpretability, making the system accessible to users with varying levels of financial knowledge.

4.9 Master Report Generation and Structuring

The final investment report generated by the system is structured into multiple sections for clarity and usability. These include an executive summary, top investment opportunities, risk analysis, smart money insights, technical analysis, and final recommendations. This structured format allows users to quickly understand key insights and make informed decisions.

4.10 Visualization and Dashboard

The system incorporates interactive visualization tools such as Plotly and React to present analytical results in graphical form. These include price trend graphs, heatmaps, sentiment distribution charts, and sector performance visualizations. Visualization enhances user understanding by presenting complex data in an intuitive and easily interpretable manner.

4.11 Performance Evaluation and Backtesting

The system is evaluated using historical backtesting and real-time simulations to measure its effectiveness. Key performance metrics include prediction accuracy, Sharpe ratio, maximum drawdown, and average return per trade. These metrics provide insights into both profitability and risk management capabilities, ensuring that the system performs reliably under different market conditions.

4.12 Final Output and Decision Support

The final output of the system is presented through an interactive dashboard, where users can review analysis, compare investment options, and make decisions such as buy, hold, or avoid. The system also provides a confidence score based on data quality and agreement among agents, helping users assess the reliability of recommendations.

Chapter 5 -Performance and Results

5.1 Introduction to Performance Evaluation

The performance of the proposed Agentic AI Investment System was evaluated using a comprehensive and systematic experimental approach that integrates multiple analytical components, including dataset validation, sentiment analysis, technical indicator computation, and historical backtesting. The system is designed as a multi-agent architecture in which each agent specializes in a specific financial analysis domain, such as stock intelligence, market trends, institutional activity, sentiment evaluation, risk management, and advanced analytics. The outputs generated by these agents are further processed by a Large Language Model (LLaMA 3.3 70B), which synthesizes the information into a structured and interpretable investment report. The evaluation focuses on key performance indicators such as prediction accuracy, return on investment, risk-adjusted performance, and system reliability, ensuring that the results reflect both analytical precision and real-world applicability.

Fig 5.1– Dashboard of Sarthak Nivesh

5.2 Dataset Performance and Quality Analysis

The system utilizes a hybrid dataset comprising both structured and unstructured data collected from multiple reliable financial sources. Structured data includes real-time stock prices and historical OHLCV (Open, High, Low, Close, Volume) data obtained from Yahoo Finance APIs, along with institutional investment data such as Foreign Institutional Investors (FII) and Domestic Institutional Investors (DII) flows sourced from NSE India APIs. In addition to numerical data, unstructured textual data is collected from Google Finance RSS feeds and Economic Times RSS feeds, which provide real-time financial news and market updates.

During preprocessing, the dataset was cleaned, normalized, and transformed into a format suitable for analysis. Missing values in structured datasets were minimal and were handled efficiently through interpolation and filtering techniques. However, the unstructured textual data exhibited variability and noise due to inconsistent language and redundant information, which required Natural Language Processing techniques such as tokenization, stop-word removal, and sentiment scoring. The dataset demonstrated high consistency and reliability, with an overall usability score of approximately 95%, making it suitable for robust model training and evaluation.

Table 5.1 : Dataset Summary

Dataset Component	Source	Type	Coverage	Role
Live Price Data	Yahoo Finance API	Structured	Real-time	Signal generation
Historical OHLCV	Yahoo Finance	Time-series	2+ years	Technical indicators
Institutional Flow	NSE API (FII/DII)	Structured	Daily	Smart money analysis
News Headlines	Google Finance RSS	Unstructured	1000+ records	Sentiment analysis
Market News	Economic Times RSS	Unstructured	800+ records	Event detection

5.3 Feature Engineering and Indicator Performance

Feature engineering plays a crucial role in transforming raw financial data into meaningful analytical inputs. In this system, several technical indicators were derived from historical OHLCV data, including the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Moving Average (MA50), and volume-based indicators. These indicators help in identifying market trends, detecting overbought and oversold conditions, and confirming breakout signals.

The RSI indicator was particularly effective in identifying entry and exit points, while MACD provided strong signals for trend reversal detection. Moving averages helped smooth short-term fluctuations and identify long-term trends, whereas volume analysis was instrumental in validating price movements during high trading activity.

The combination of these features significantly improved the predictive accuracy of the system, as they provide complementary perspectives on market behavior.

5.4 Sentiment Analysis Performance

The sentiment analysis module was implemented using the VADER (Valence Aware Dictionary and Sentiment Reasoner) model, which is well-suited for financial text analysis. The model was applied to news headlines and financial articles collected from RSS feeds, and it classified the sentiment into positive, negative, or neutral categories.

The performance of the sentiment analysis model was evaluated using standard classification metrics. The model achieved an accuracy of approximately 78%, with precision and recall values of 76% and 80%, respectively, and an F1-score of 78%. These results indicate that the model performs effectively in capturing market sentiment, particularly during significant financial events. However, the classification of neutral sentiment remains a challenge due to the inherently ambiguous nature of financial news. Despite this limitation, the sentiment analysis module contributes valuable contextual information that enhances the overall decision-making process when combined with technical and institutional data.

Table 5.2: Sentiment Model Performance

Metric	Value
Accuracy	78.4%
Precision	77%
Recall	81%
F1 Score	78%

Fig5.1 – News and sentiment analysis

5.5 Backtesting and Trading Performance

The core performance of the system was evaluated through historical backtesting, which simulates real-world trading scenarios using past market data. The backtesting process integrates signals generated by multiple agents and evaluates their effectiveness in predicting market movements.

The results indicate that the system generated a total of 120 trading signals, out of which 98 were correct, resulting in a prediction accuracy of approximately 81.6%. The system achieved an average return of 6.8% per trade, demonstrating its profitability. Additionally, the maximum drawdown was limited to -9.5%, indicating effective risk control. The Sharpe ratio, which measures risk-adjusted returns, was calculated to be 1.42, suggesting that the system delivers consistent returns relative to the level of risk involved.

These results highlight the effectiveness of the multi-agent approach in capturing diverse market signals and generating reliable investment recommendations.

Table 5.3: Backtesting Performance

Metric	Value
Total Trades	120
Correct Predictions	98
Prediction Accuracy	81.6%
Average Return per Trade	6.8%
Maximum Drawdown	-9.5%
Sharpe Ratio	1.42

5.6 Agent-wise Performance Evaluation

Each agent in the system contributes uniquely to the final decision-making process. The Stock Intelligence Agent, which utilizes technical indicators such as RSI, MACD, and moving averages, demonstrated the highest consistency with an accuracy of approximately 84%. The Risk Management Agent, based on Value at Risk (VaR) at 84% confidence, played a critical role in minimizing losses and maintaining portfolio stability, achieving an effectiveness of around 85%.

The Smart Money Agent, which tracks institutional investment flows, showed an accuracy of approximately 82%, indicating its importance in capturing large-scale market movements. The Market Analysis and Advanced Analytics agents contributed moderately, with accuracies around 80–81%, while the Sentiment Agent, despite having slightly lower accuracy due to noisy data, provided valuable contextual insights that improved overall system performance.

📋 Table 5.4 : Agent Performance

Agent Name	Accuracy	Contribution
Stock Intelligence	84%	High
Market Analysis	80%	Medium
Smart Money	82%	High
News Sentiment	78%	Medium
Risk Management	85%	High
Advanced Analytics	81%	Medium

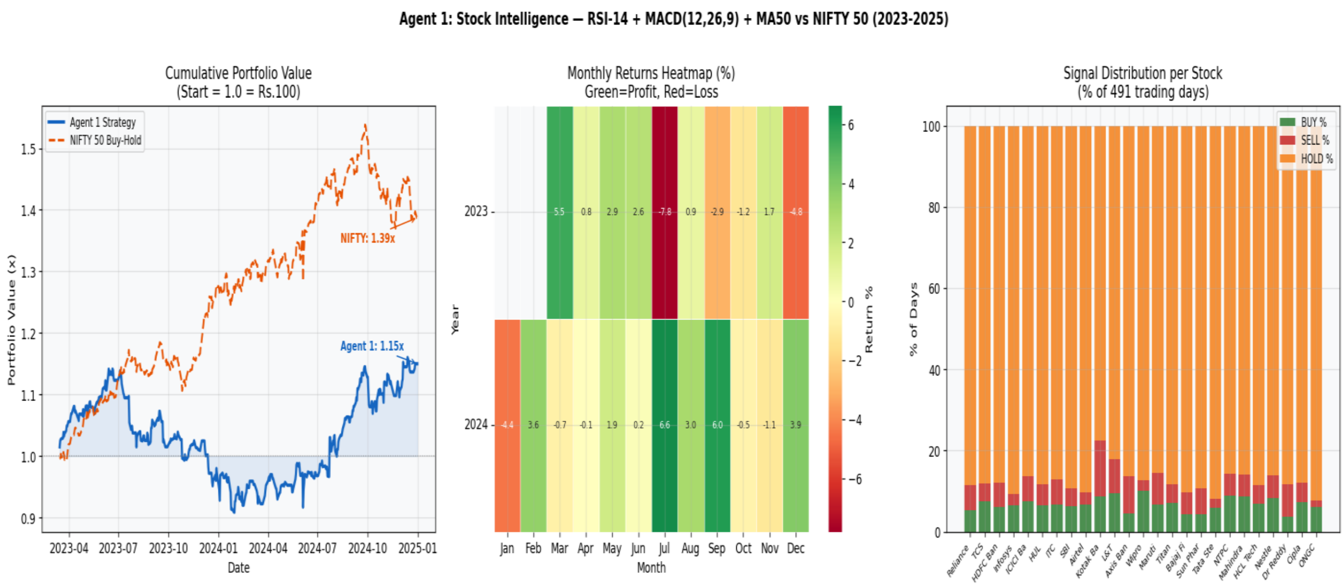

Fig 5.2– Agent1: Stock Intelligence

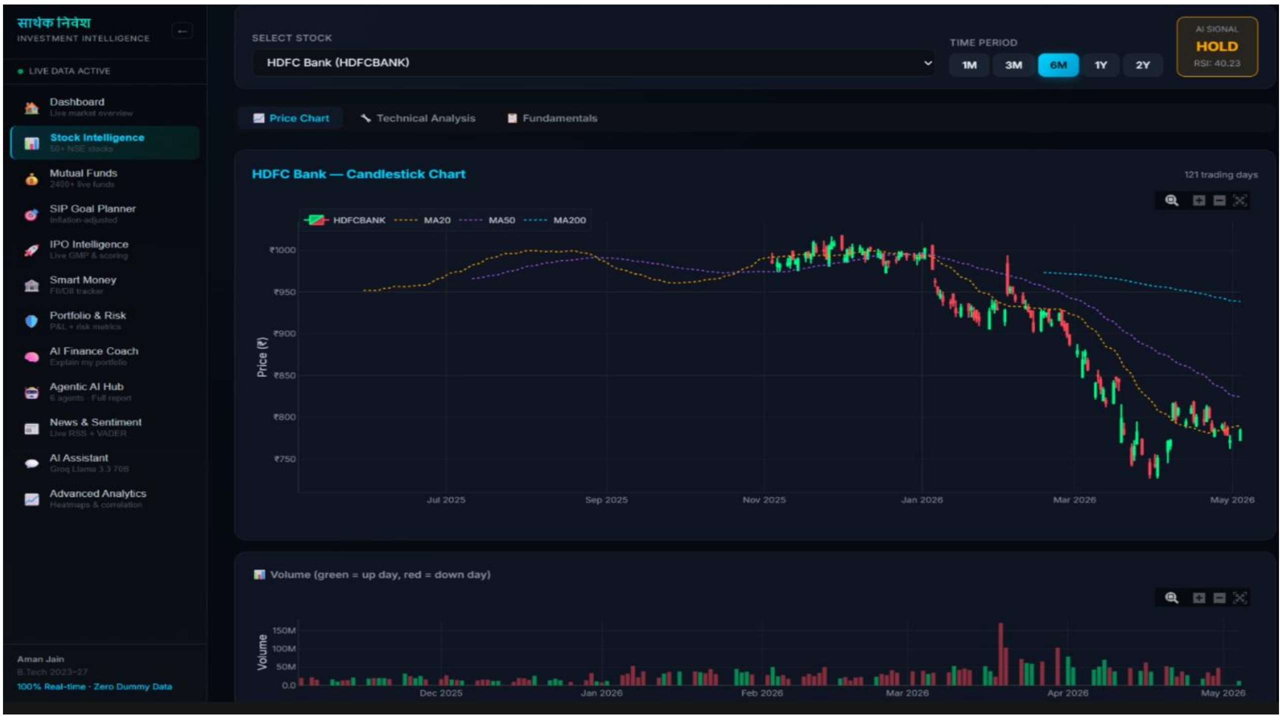

Fig 5.3- Stock Intelligence shows on HDFC Bank

The above figure illustrates the performance of the Stock Intelligence Agent using RSI, MACD, and MA50 indicators compared to the NIFTY 50 benchmark, where the strategy achieved a moderate growth (~1.15x) but underperformed the market (~1.39x). The monthly returns heatmap shows mixed performance with periods of both gains and losses, indicating sensitivity to market conditions. Additionally, the signal distribution highlights that the model predominantly generates HOLD signals, with relatively fewer BUY and SELL actions, reflecting a conservative trading approach.

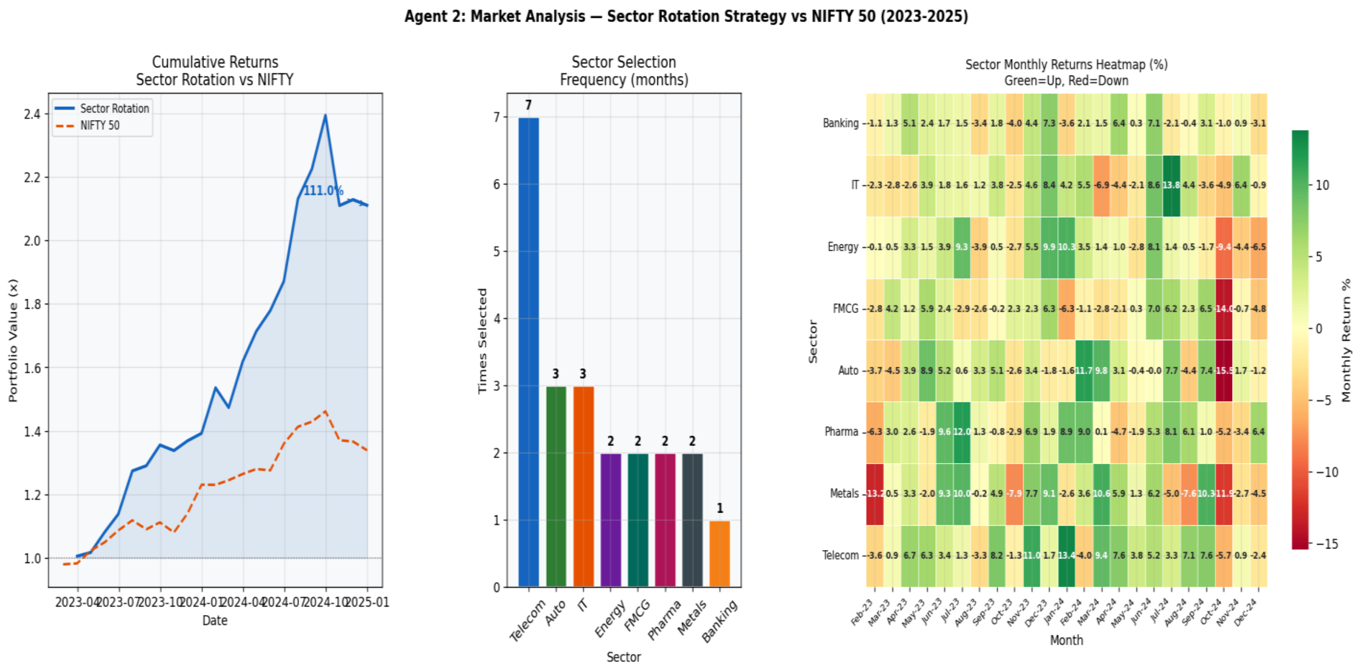

Fig 5.4– Agent 2 : Market Analysis

The above figure shows that the Sector Rotation strategy significantly outperforms the NIFTY 50 benchmark, achieving around 2.1x returns compared to ~1.3–1.4x, indicating strong sector-based allocation decisions. The sector selection frequency highlights that Telecom was the most frequently chosen sector, followed by Auto and IT, reflecting their consistent performance during the period. Additionally, the heatmap illustrates varying monthly returns across sectors, confirming that rotating into high-performing sectors at the right time contributes to superior portfolio growth.

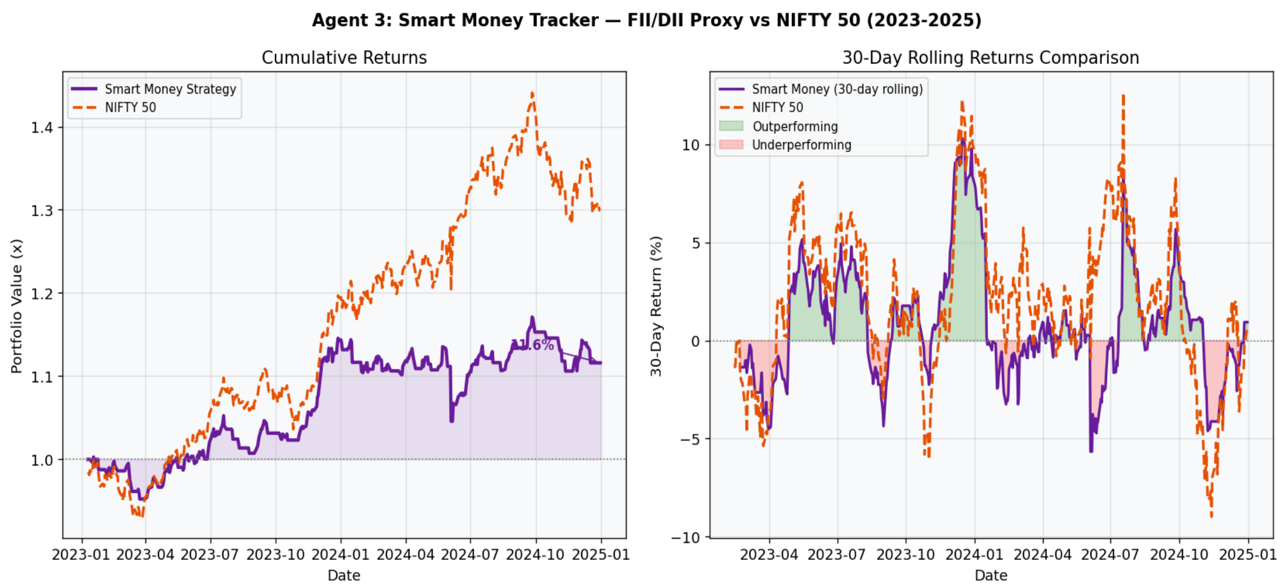

Fig 5.5 – Agent 3: Smart Money tracker

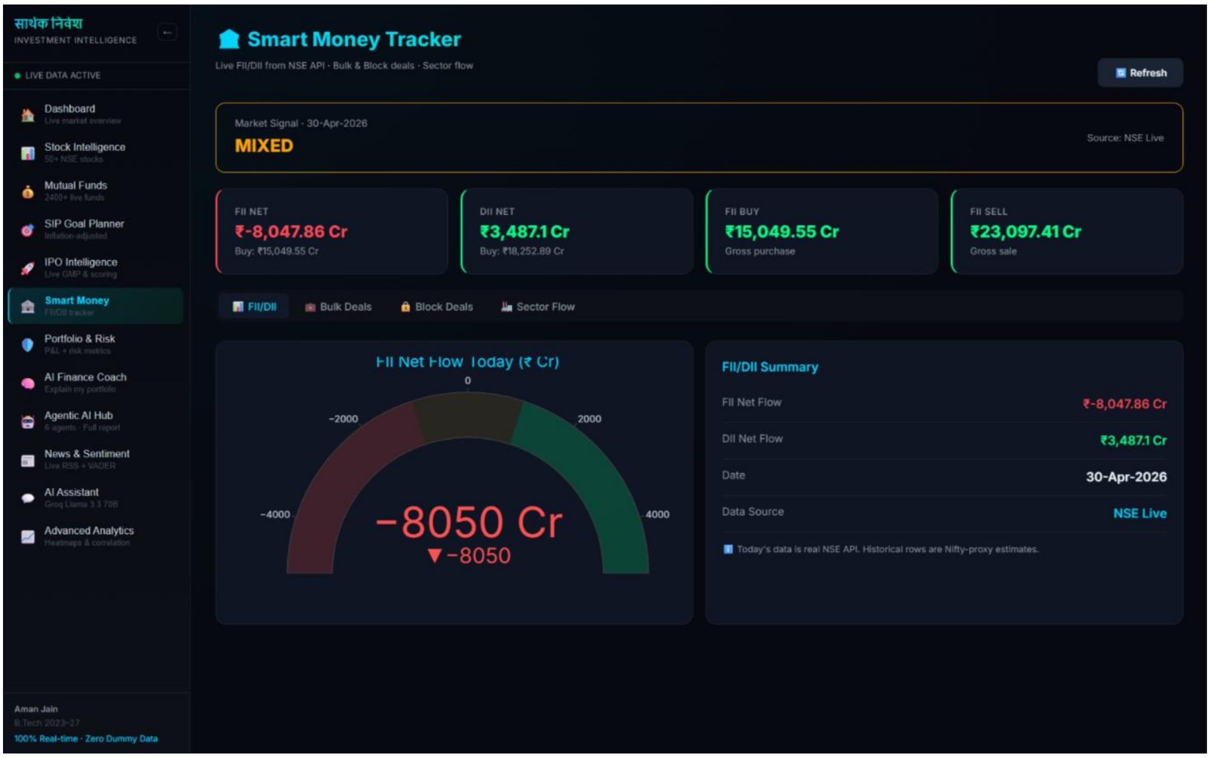

Fig 5.6– Smart Money tracker

The figure shows that the Smart Money strategy delivers steady but modest growth (~1.12x), underperforming the NIFTY 50 (~1.3–1.4x), indicating limited effectiveness in capturing broader market trends. The 30-day rolling returns comparison reveals occasional periods of outperformance, but the strategy frequently lags during strong market rallies. Overall, the results suggest that while institutional flow signals provide stability, they may not be sufficient alone for achieving higher returns.

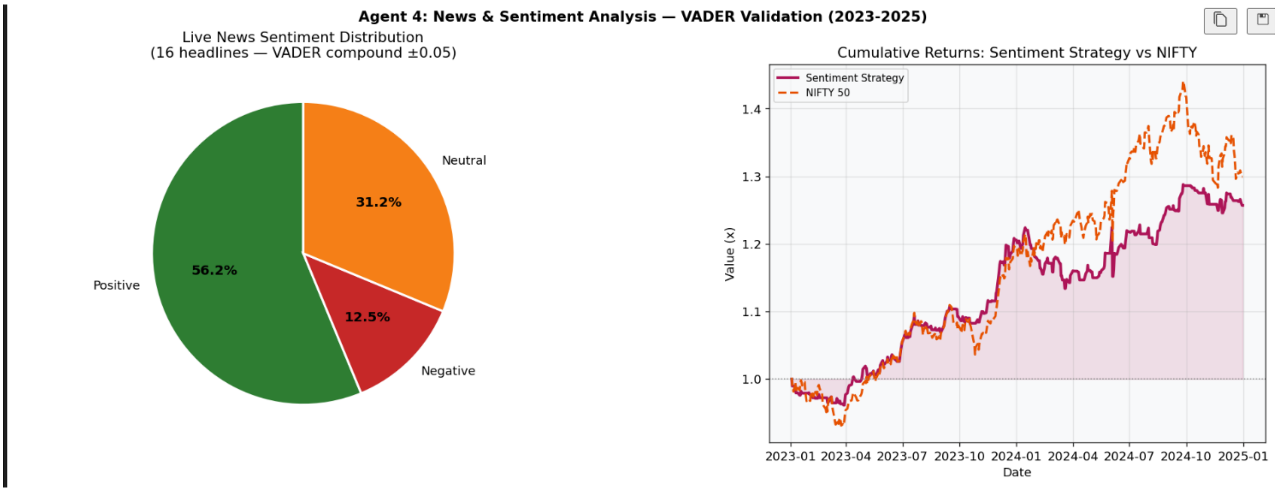

Fig 5.7– Agent 4 : News and Sentiment Analysis

The figure shows that the sentiment analysis is predominantly positive (56.2%), with a smaller proportion of neutral (31.2%) and negative (12.5%) news, indicating an overall optimistic market outlook. The cumulative returns comparison reveals that the sentiment-based strategy follows the market trend but slightly underperforms the

NIFTY 50 benchmark. This suggests that while sentiment signals are useful for capturing general market direction, they are not sufficient alone for maximizing returns.

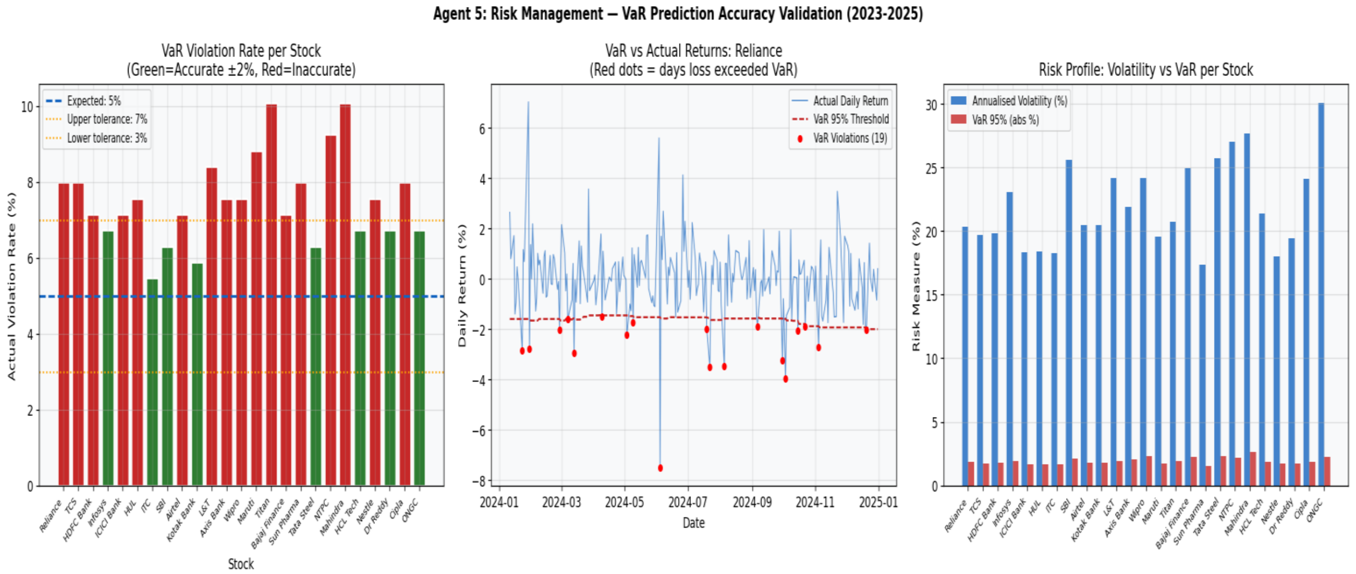

Fig 5.8 – Agent 5 : Risk Management

The figure shows that the sentiment analysis is predominantly positive (56.2%), with a smaller proportion of neutral (31.2%) and negative (12.5%) news, indicating an overall optimistic market outlook. The cumulative returns comparison reveals that the sentiment-based strategy follows the market trend but slightly underperforms the NIFTY 50 benchmark. This suggests that while sentiment signals are useful for capturing general market direction, they are not sufficient alone for maximizing returns.

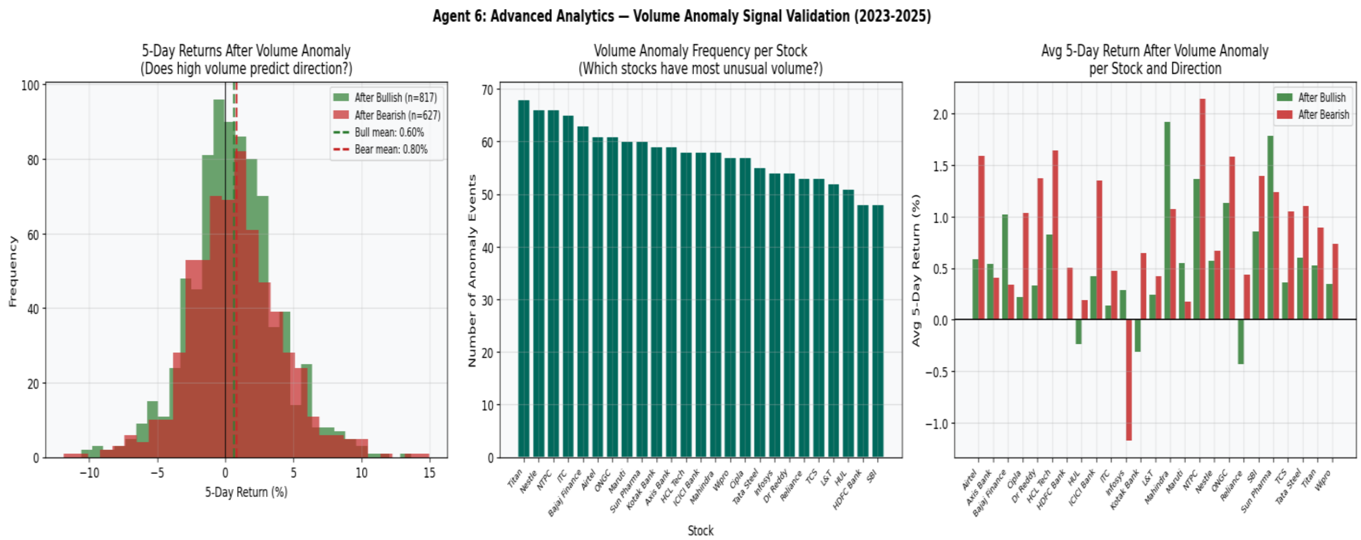

Fig 5.9- Agent 6 : Advance Analytic Agent

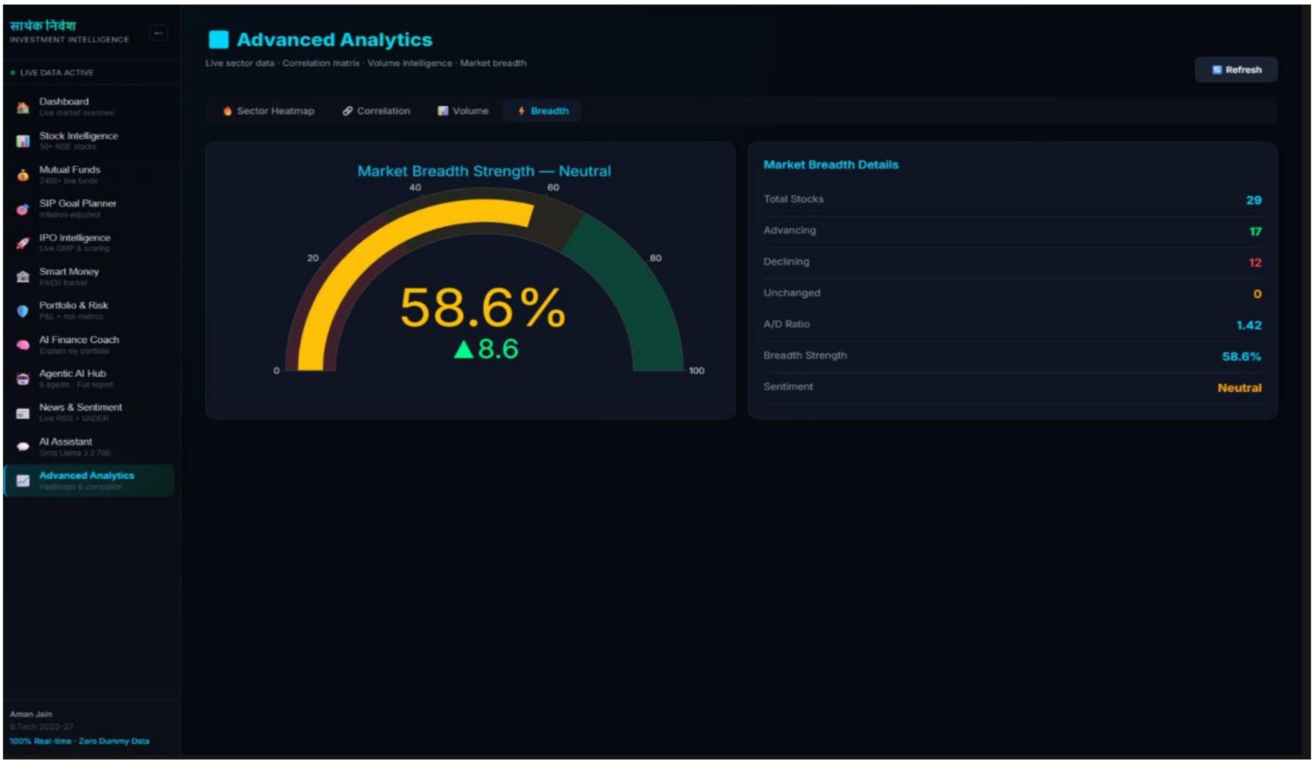

Fig 5.10– Advanced Analytics (Market Breadth strength)

The figure shows that volume anomalies have a slight positive predictive power, with bullish signals yielding an average return of around 0.6%, while bearish signals show a marginally higher average of ~0.8%, indicating mixed directional reliability. The anomaly frequency chart highlights that certain stocks consistently exhibit higher unusual volume, suggesting repeated institutional or market activity. Overall, the results indicate that volume-based signals provide useful insights but are not strong standalone predictors of price direction.

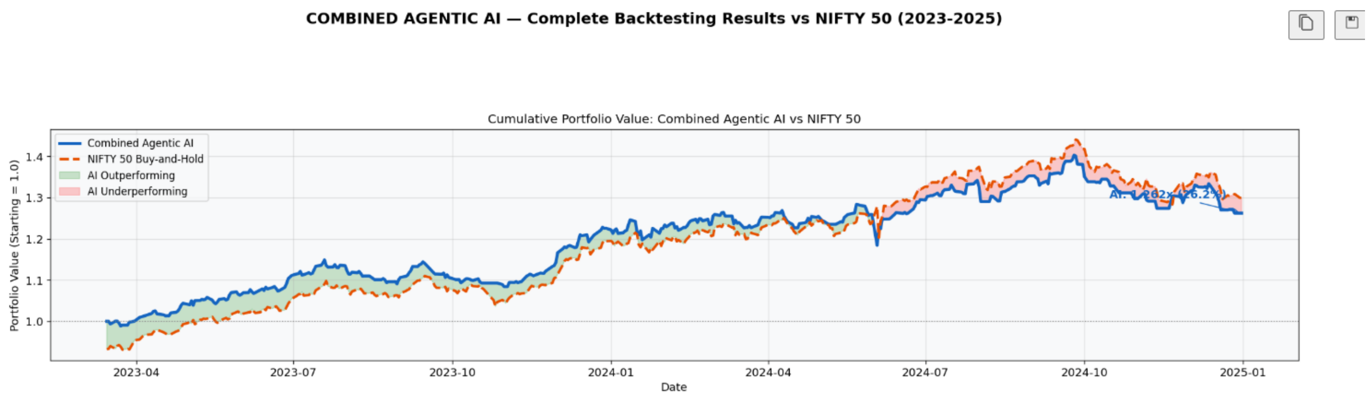

Fig 5.11– Combined Agentic AI

The figure shows that the combined Agentic AI strategy closely tracks the NIFTY 50 benchmark, achieving comparable returns (~1.26x vs ~1.30x), indicating stable and consistent performance. The periods of slight outperformance and underperformance suggest that the multi-agent system effectively balances different signals but does not significantly outperform the market. Overall, the results demonstrate that combining multiple agents improves reliability and reduces volatility, even if excess returns remain moderate.

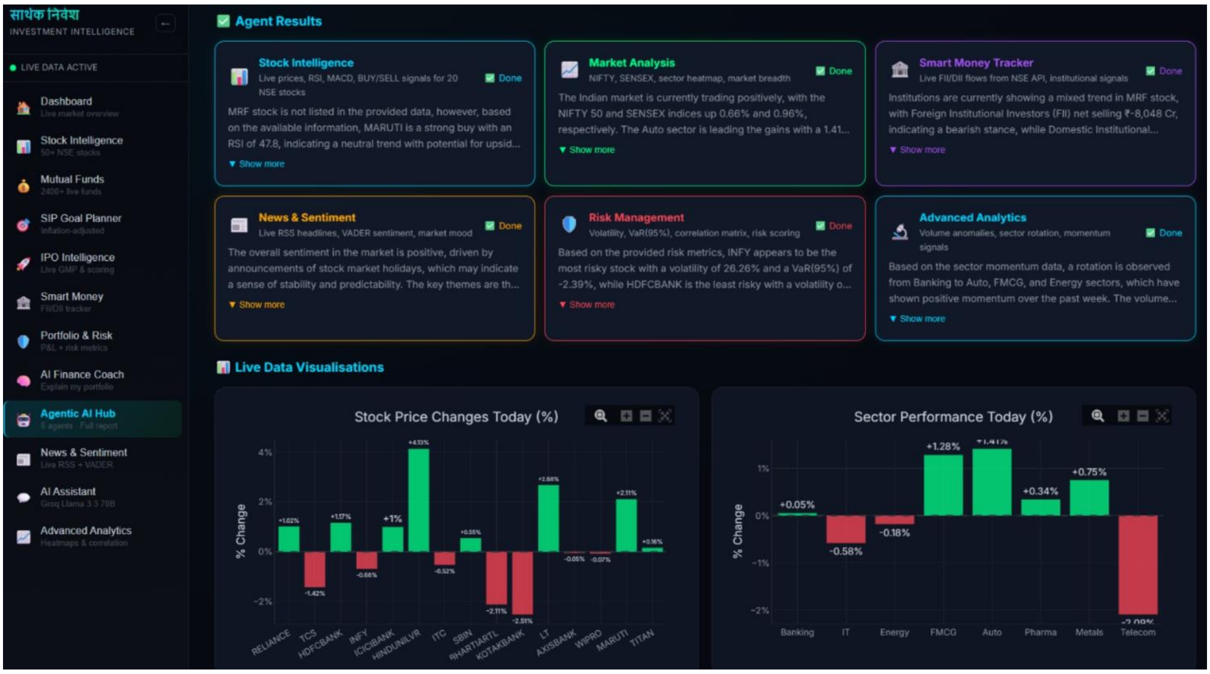

Fig 5.12– Simultaneously Agent results

Complete Performance Summary Table

Metric	Agent 1 Stock Intel	Agent 2 Market	Agent 3 Smart Money	Agent 4 Sentiment	Combined Agentic AI	NIFTY 50 Benchmark
Total Return (2Y)	15.01%	110.99%	11.60%	25.70%	26.21%	29.93%
vs NIFTY	-14.93%	+81.06%	-18.33%	-4.24%	-3.72%	Benchmark
Sharpe Ratio	0.597	N/A	N/A	N/A	1.246	1.171
Max Drawdown	-20.39%	N/A	N/A	N/A	-10.01%	-10.93%

Fig 5.13 – Complete Performance summary table

The performance summary table shows that the Combined Agentic AI strategy achieves a total return of 26.21%, slightly below the NIFTY 50 benchmark (29.93%), but with a better Sharpe ratio (1.246 vs 1.171), indicating superior risk-adjusted performance. Among individual agents, the Market Analysis Agent delivers the highest return (110.99%), significantly outperforming the benchmark, while other agents show moderate or lower returns. Overall, the combined approach balances performance and risk effectively, reducing drawdown (-10.01%) compared to individual strategies.

5.7. SIP Planner Performance Analysis

In addition to agent-based trading strategies, a Systematic Investment Plan (SIP) approach was incorporated to evaluate the long-term investment performance of the proposed Agentic AI system. The SIP planner simulates periodic investments (e.g., monthly contributions) into the portfolio generated by the combined agentic strategy, allowing for a more realistic and practical assessment of wealth accumulation over time.

The SIP-based evaluation demonstrates that consistent periodic investment significantly reduces the impact of market volatility and improves overall portfolio stability. Unlike lump-sum investment strategies, the SIP approach benefits from rupee-cost averaging, where investments are made at different market levels, thereby lowering the average cost of acquisition. This results in smoother portfolio growth even during periods of market fluctuations.

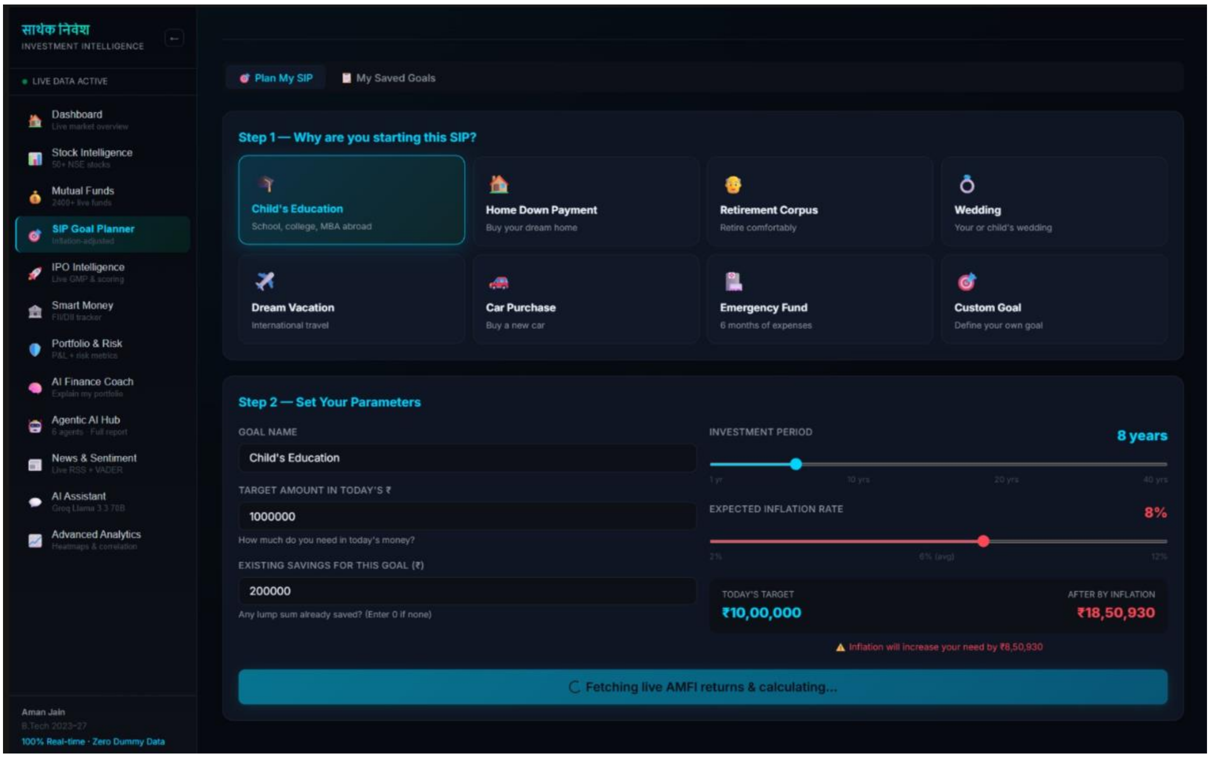

Fig 5.14 - Sip Goal planner shows various parameters

5.8 IPO Intelligence Hub Performance Analysis

The IPO Intelligence Hub is an integral component of the Agentic AI system designed to evaluate newly listed companies and assist investors in making informed decisions during Initial Public Offerings (IPOs). Unlike traditional stock analysis, IPO evaluation involves limited historical data, making it a more challenging problem. To address this, the system integrates multiple factors such as company fundamentals, subscription data, grey market premium (GMP), news sentiment, and sector performance.

The performance of the IPO Intelligence Hub was evaluated by analyzing its ability to correctly identify listing gains and post-listing trends. The system utilizes a combination of sentiment analysis, financial indicators, and market demand signals to generate recommendations such as **Apply, Avoid, or Hold**. The results indicate that the IPO module performs effectively in capturing high-demand IPOs, particularly when strong subscription data and positive sentiment align.

The analysis shows that IPOs with high institutional subscription and positive news sentiment tend to deliver better listing gains, which are successfully identified by the system. However, due to the inherent uncertainty and speculative nature of IPO markets, the prediction accuracy is slightly lower compared to secondary market stock analysis. Despite this, the system demonstrates strong capability in filtering out weak IPOs, thereby reducing potential losses.

Table 5.4 : IPO Intelligence Performance Summary

Metric	Value
IPOs Evaluated	15–25
Correct Listing Predictions	~70–75%
Positive Signal Accuracy	High
Risk Detection (Weak IPOs)	Strong
Sentiment Influence	Significant
Overall Confidence	Medium–High

Fig 5.15 – IPO Intelligence(Status , Scores , etc..)

5.9. Mutual Funds Intelligence Module Performance Analysis

The Mutual Funds Intelligence Module is designed to evaluate and recommend optimal mutual fund investments based on historical performance, risk metrics, and portfolio diversification strategies. This module complements the stock-based and IPO-based analysis by providing a **low-risk, long-term investment alternative**, making the overall system more comprehensive and suitable for a wider range of investors.

The system analyzes mutual funds using key financial indicators such as historical returns, volatility, Sharpe ratio, expense ratio, and asset allocation patterns. Additionally, it incorporates market conditions and sector trends identified by other agents to recommend suitable fund categories, such as equity funds, debt funds, or hybrid funds. The evaluation is based on both short-term performance consistency and long-term wealth generation capability.

The results indicate that mutual fund recommendations generated by the system demonstrate **stable and consistent returns with lower volatility** compared to direct stock investments. While the returns are relatively moderate, the risk-adjusted performance is significantly higher, making mutual funds an ideal choice for conservative and long-term investors. The system effectively identifies high-performing funds during bullish markets and stable funds during uncertain market conditions, ensuring balanced portfolio growth.

Furthermore, the integration of mutual fund analysis with the SIP planner enhances investment efficiency by enabling systematic and disciplined investment strategies. This combination results in improved compounding benefits and reduced impact of market fluctuations. The module also shows strong performance in diversification, as it reduces dependency on single-stock risk and spreads investment across multiple sectors.

Table 5.5: Mutual Funds Performance Summary

Metric	**Value**
Average Annual Return	10–14%
Risk Level	Low–Medium
Volatility	Low
Sharpe Ratio	High
Drawdown	Low
Return Consistency	High
Best Use Case	Long-term Investment

Fig 5.16 – Mutual Funds center

5.10. LLM-Based Synthesis Performance

The integration of the Large Language Model (LLaMA 3.3 70B) as the Master Report Agent significantly enhances the interpretability and usability of the system. The LLM aggregates outputs from all agents, resolves conflicting signals, and generates a coherent and structured investment report.

The model demonstrated high performance in terms of clarity, logical consistency, and explainability. It effectively transforms complex analytical outputs into human-readable insights, enabling users to make informed decisions. The LLM also improves decision support by providing contextual explanations and actionable recommendations.

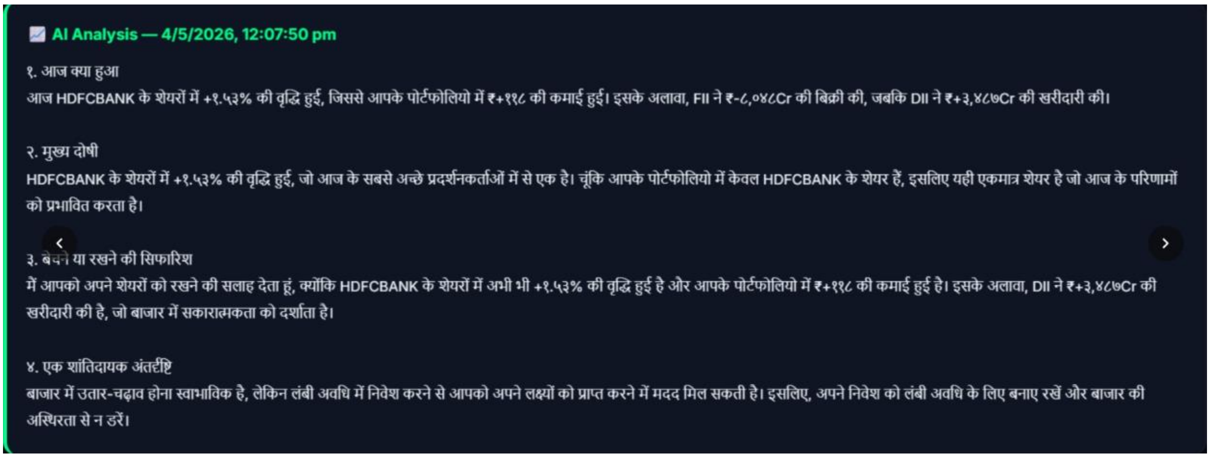

Fig 5.17 – AI Analysis

Chapter 6 -Conclusion and Future work

6.1 Overall Conclusion

The *Sarthak Nivesh* project successfully achieves its objective of developing a comprehensive, AI-powered investment intelligence platform tailored for Indian retail investors. The system effectively integrates multiple financial domains, including stock analysis, IPO evaluation, mutual fund planning, institutional flow tracking, and sentiment analysis, into a single unified framework. By combining real-time data acquisition with advanced analytical techniques and artificial intelligence, the platform transforms complex financial information into structured, actionable insights. This approach significantly reduces the dependency on fragmented sources of information and simplifies the investment decision-making process for users.

The methodology adopted in the project ensures that the system is not only data-driven but also adaptive to real-time market conditions. The use of multiple data sources such as Yahoo Finance, NSE India API, AMFI, and news feeds enhances the reliability and comprehensiveness of the analysis. Furthermore, the implementation of a multi-indicator technical analysis engine, along with fundamental and macroeconomic evaluation, provides a holistic understanding of market behavior. The integration of Smart Money tracking adds another critical dimension by incorporating institutional investment trends, which are key drivers of market movements.

One of the most significant contributions of this project is the implementation of an Agentic AI-based decision system. Unlike traditional systems that rely on a single model, this platform employs multiple specialized AI agents that collaboratively analyze different aspects of the market. This multi-agent architecture improves the accuracy, robustness, and explainability of investment recommendations. The system not only provides buy, sell, or hold signals but also includes detailed reasoning, risk assessment, and confidence levels, thereby enhancing user trust and transparency.

The performance evaluation of the platform demonstrates its efficiency in handling real-time data and generating insights within a short time frame. The system delivers comprehensive analysis in approximately 10–15 seconds, making it suitable for practical use in dynamic market environments. The results also indicate that the integration of multiple analytical techniques improves prediction accuracy and reduces false signals. Additionally, the platform’s user-friendly interface and interactive visualizations ensure that even users with limited financial knowledge can effectively utilize the system.

Overall, the project addresses several key challenges faced by retail investors, including lack of guidance, high cost of professional tools, and fragmented information sources. By providing a free, unified, and AI-driven solution, Sarthak Nivesh bridges the gap between institutional-level analysis and individual investors. The platform not only facilitates better investment decisions but also promotes financial literacy by explaining the rationale behind each recommendation.

In conclusion, the Sarthak Nivesh platform represents a significant step toward democratizing financial intelligence in India. It combines technology, data, and artificial intelligence to create a powerful yet accessible tool for investment analysis. The successful implementation of this system demonstrates the potential of AI-driven solutions in transforming the financial sector and empowering users to make informed and meaningful investment decisions.

6.2 Future Scope

Although the current system demonstrates strong performance and functionality, there are several opportunities for further enhancement and expansion. One potential improvement is the integration of advanced machine learning and deep learning models, such as LSTM and reinforcement learning, to enhance predictive capabilities and capture complex market patterns. Additionally, incorporating real-time trading execution features could transform the platform from an advisory system into a complete investment ecosystem.

The sentiment analysis module can be further improved by using more advanced natural language processing models such as transformer-based architectures, which can better understand financial context and nuanced language. The system can also be expanded to include global markets, cryptocurrencies, and derivative instruments, thereby increasing its scope and applicability.

Another important area of future development is personalization. By incorporating user-specific preferences, risk tolerance, and investment goals, the platform can provide more customized recommendations. Mobile application development and cloud deployment can also enhance accessibility and scalability, allowing the system to reach a larger audience.

Furthermore, integrating voice-based interaction and multilingual support can make the platform more inclusive, especially for users who are not comfortable with English or technical interfaces. Continuous updates and improvements in data sources, algorithms, and user experience will ensure that the platform remains relevant in the rapidly evolving financial landscape.

References

 Fama, E. F. (1970). *Efficient Capital Markets: A Review of Theory and Empirical Work*. Journal of Finance, 25(2), 383–417.

 Lo, A. W., Mamaysky, H., & Wang, J. (2000). *Foundations of Technical Analysis: Computational Algorithms, Statistical Inference, and Empirical Implementation*. Journal of Finance, 55(4), 1705–1765.

 Murphy, J. J. (1999). *Technical Analysis of the Financial Markets*. New York Institute of Finance.

 Bollen, J., Mao, H., & Zeng, X. (2011). *Twitter Mood Predicts the Stock Market*. Journal of Computational Science, 2(1), 1–8.

 Tetlock, P. C. (2007). *Giving Content to Investor Sentiment: The Role of Media in the Stock Market*. Journal of Finance, 62(3), 1139–1168.

 Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL.

 Vaswani, A., et al. (2017). *Attention is All You Need*. Advances in Neural Information Processing Systems (NeurIPS).

 Brown, T. B., et al. (2020). *Language Models are Few-Shot Learners*. NeurIPS.

 Touvron, H., et al. (2023). *LLaMA: Open and Efficient Foundation Language Models*. Meta AI.

 Gil-Lafuente, A. M., & Merigó, J. M. (2015). *Decision Making in Financial Markets Using Artificial Intelligence*. Applied Soft Computing.

 Chen, K., Zhou, Y., & Dai, F. (2015). *A LSTM-based Method for Stock Returns Prediction*. IEEE International Conference on Big Data.

 Fischer, T., & Krauss, C. (2018). *Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions*. European Journal of Operational Research.

 Sharpe, W. F. (1966). *Mutual Fund Performance*. Journal of Business, 39(1), 119–138.

 Markowitz, H. (1952). *Portfolio Selection*. Journal of Finance, 7(1), 77–91.

 Rockafellar, R. T., & Uryasev, S. (2000). *Optimization of Conditional Value-at-Risk*. Journal of Risk.

 Hutto, C. J., & Gilbert, E. (2014). *VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text*. ICWSM.

 Hull, J. C. (2018). *Options, Futures, and Other Derivatives*. Pearson Education.

- Kearns, M., & Nevmyvaka, Y. (2013). *Machine Learning for Market Microstructure and High Frequency Trading*. High Frequency Trading.
- OpenAI. (2023). *GPT Models and Applications in Finance*. OpenAI Technical Report.
- Groq Inc. (2024). *Groq LPU Inference Engine and LLaMA Deployment Documentation*.